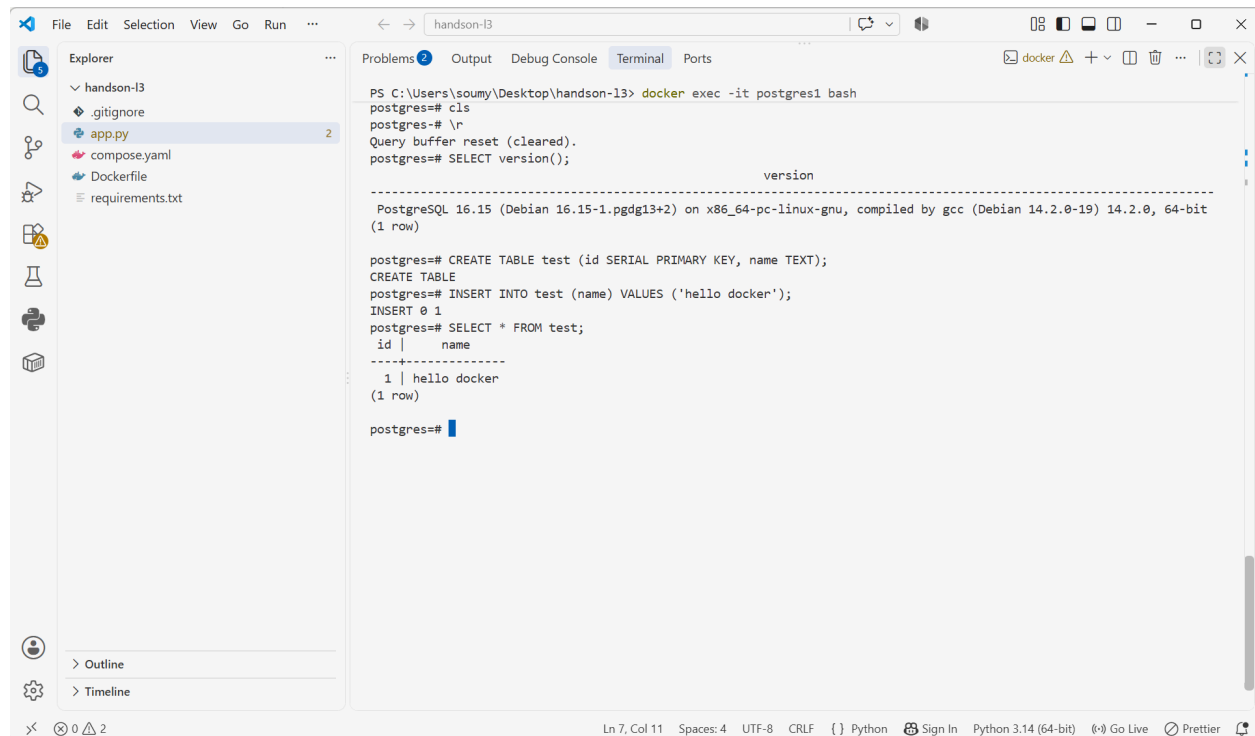


Hands-On L3: Installing Docker & Building a Multi-Container Microservice

Soumyanil Ain — ITCS 6190/8190, Cloud Computing for Data Analysis — Fall 2026

GitHub repository: <https://github.com/soumyanilain/handson-l3-SoumyanilAin>

Figure 1 — psql session inside the postgres1 container, showing SELECT version() and a test table query.

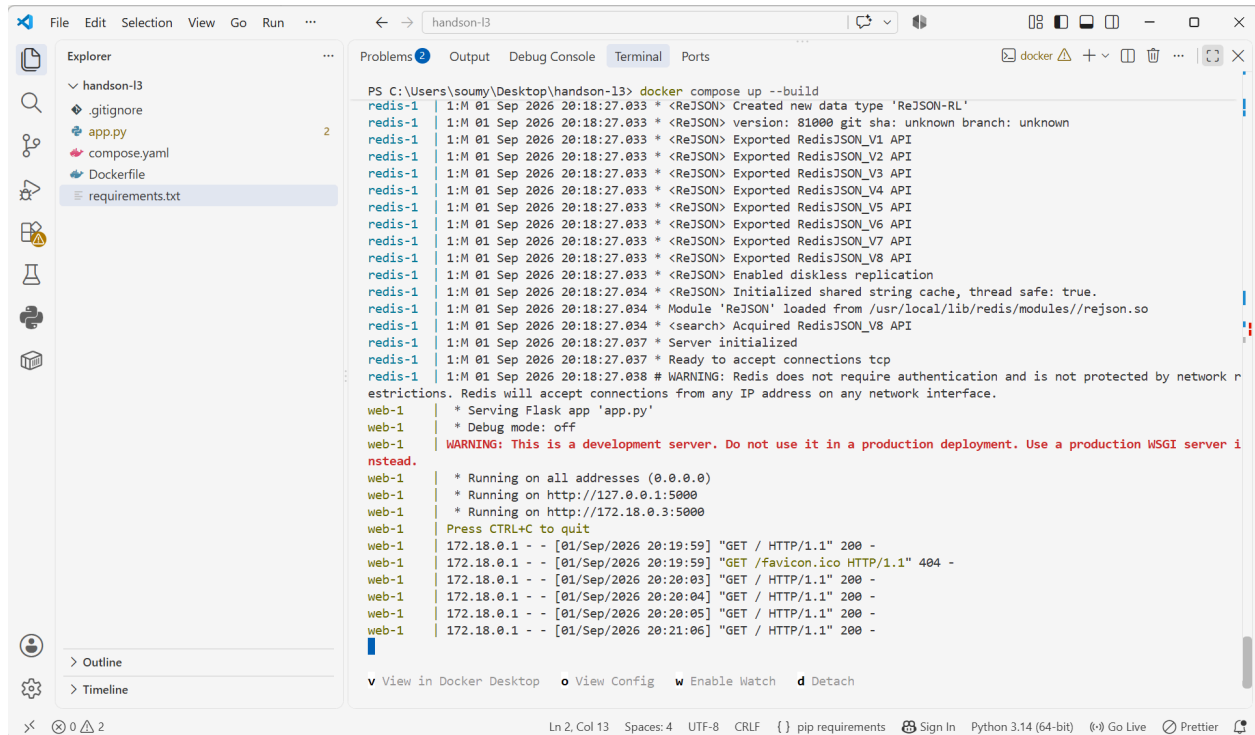


```
PS C:\Users\soumy\Desktop\handson-l3> docker exec -it postgres1 bash
postgres=# \c
postgres=# \r
Query buffer reset (cleared).
postgres=# SELECT version();
               version
-----
PostgreSQL 16.15 (Debian 16.15-1.pgdg13+2) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
(1 row)

postgres=# CREATE TABLE test (id SERIAL PRIMARY KEY, name TEXT);
CREATE TABLE
postgres=# INSERT INTO test (name) VALUES ('hello docker');
INSERT 0 1
postgres=# SELECT * FROM test;
 id | name
----+-----
  1 | hello docker
(1 row)

postgres=#
```

Figure 2 — Docker Desktop Containers view showing the handson-l3 stack with web-1 and redis-1 both running.



```
PS C:\Users\soumy\Desktop\handson-13> docker compose up --build
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Created new data type 'ReJSON-RL'
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> version: 81000 git sha: unknown branch: unknown
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V1 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V2 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V3 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V4 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V5 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V6 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V7 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Exported RedisJSON_V8 API
redis-1 1:M 01 Sep 2026 20:18:27.033 * <ReJSON> Enabled diskless replication
redis-1 1:M 01 Sep 2026 20:18:27.034 * <ReJSON> Initialized shared string cache, thread safe: true.
redis-1 1:M 01 Sep 2026 20:18:27.034 * Module 'ReJSON' loaded from /usr/local/lib/redis/modules/rejson.so
redis-1 1:M 01 Sep 2026 20:18:27.034 * <search> Acquired RedisJSON_V8 API
redis-1 1:M 01 Sep 2026 20:18:27.037 * Server initialized
redis-1 1:M 01 Sep 2026 20:18:27.037 * Ready to accept connections tcp
redis-1 1:M 01 Sep 2026 20:18:27.038 # WARNING: Redis does not require authentication and is not protected by network restrictions. Redis will accept connections from any IP address on any network interface.
web-1   | * Serving Flask app 'app.py'
web-1   | * Debug mode: off
web-1   | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
web-1   | * Running on all addresses (0.0.0.0)
web-1   | * Running on http://127.0.0.1:5000
web-1   | * Running on http://172.18.0.3:5000
web-1   | Press CTRL+C to quit
web-1   | 172.18.0.1 - - [01/Sep/2026 20:19:59] "GET / HTTP/1.1" 200 -
web-1   | 172.18.0.1 - - [01/Sep/2026 20:19:59] "GET /favicon.ico HTTP/1.1" 404 -
web-1   | 172.18.0.1 - - [01/Sep/2026 20:20:03] "GET / HTTP/1.1" 200 -
web-1   | 172.18.0.1 - - [01/Sep/2026 20:20:04] "GET / HTTP/1.1" 200 -
web-1   | 172.18.0.1 - - [01/Sep/2026 20:20:05] "GET / HTTP/1.1" 200 -
web-1   | 172.18.0.1 - - [01/Sep/2026 20:21:06] "GET / HTTP/1.1" 200 -
```

Figure 3 — `http://localhost:8000` after several refreshes; the counter increments because Redis persists the hits key across requests.



Figure 4 — Docker Desktop Containers view showing the handson-l3 Compose stack with both services running: redis-1 (redis:alpine) and web-1 (handson-l3, published on 8000:5000).

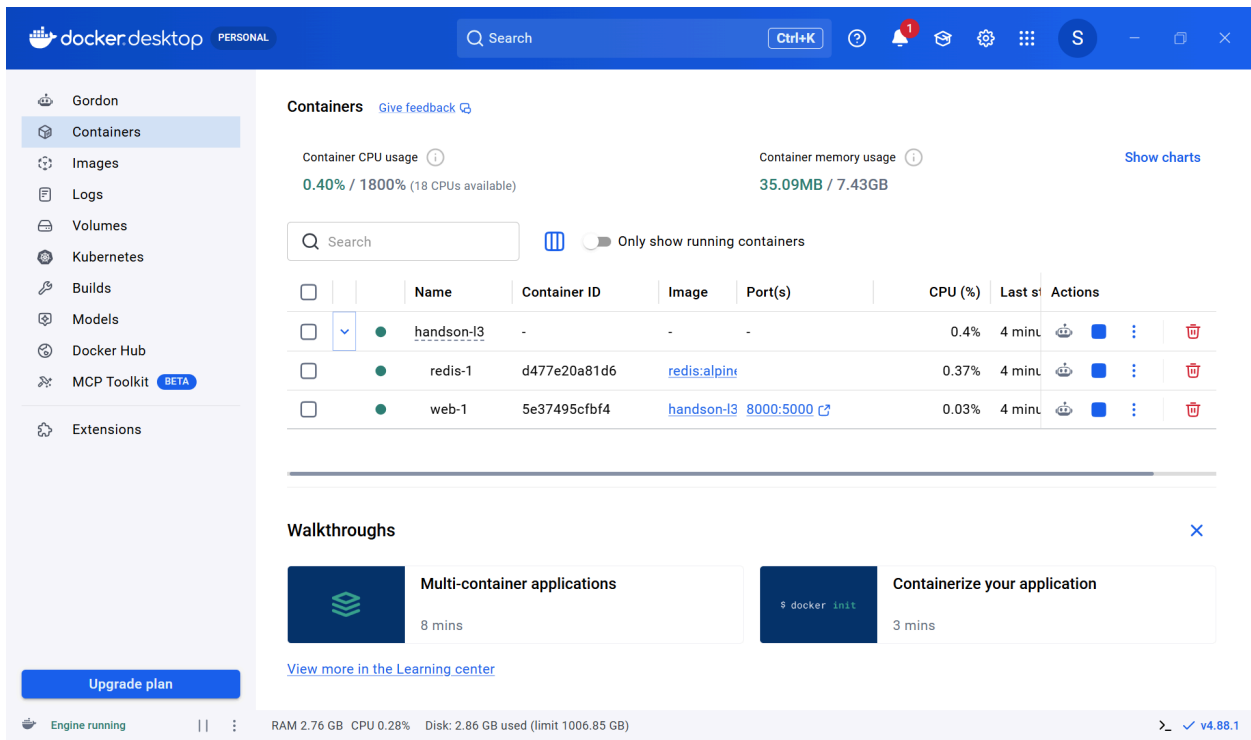


Figure 5 — Container logs for handson-l3-web, showing the Flask server starting on 172.18.0.3:5000 and successive GET / requests returning 200 as the page was refreshed.

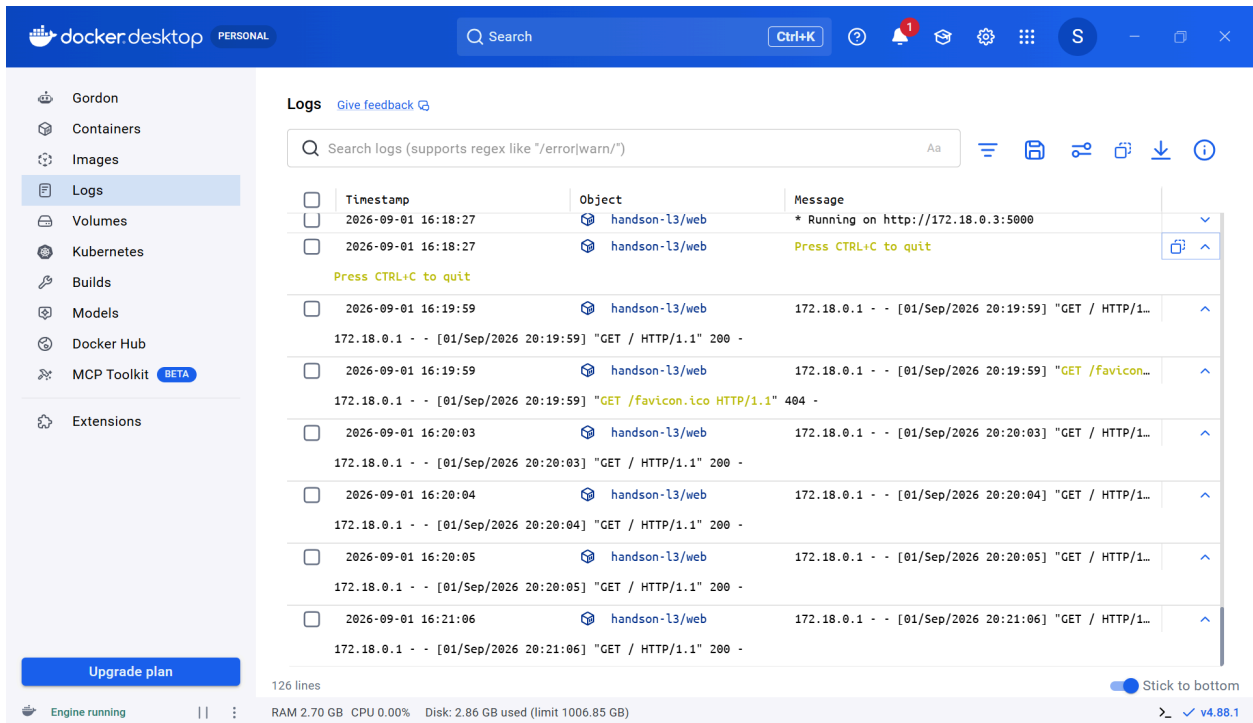
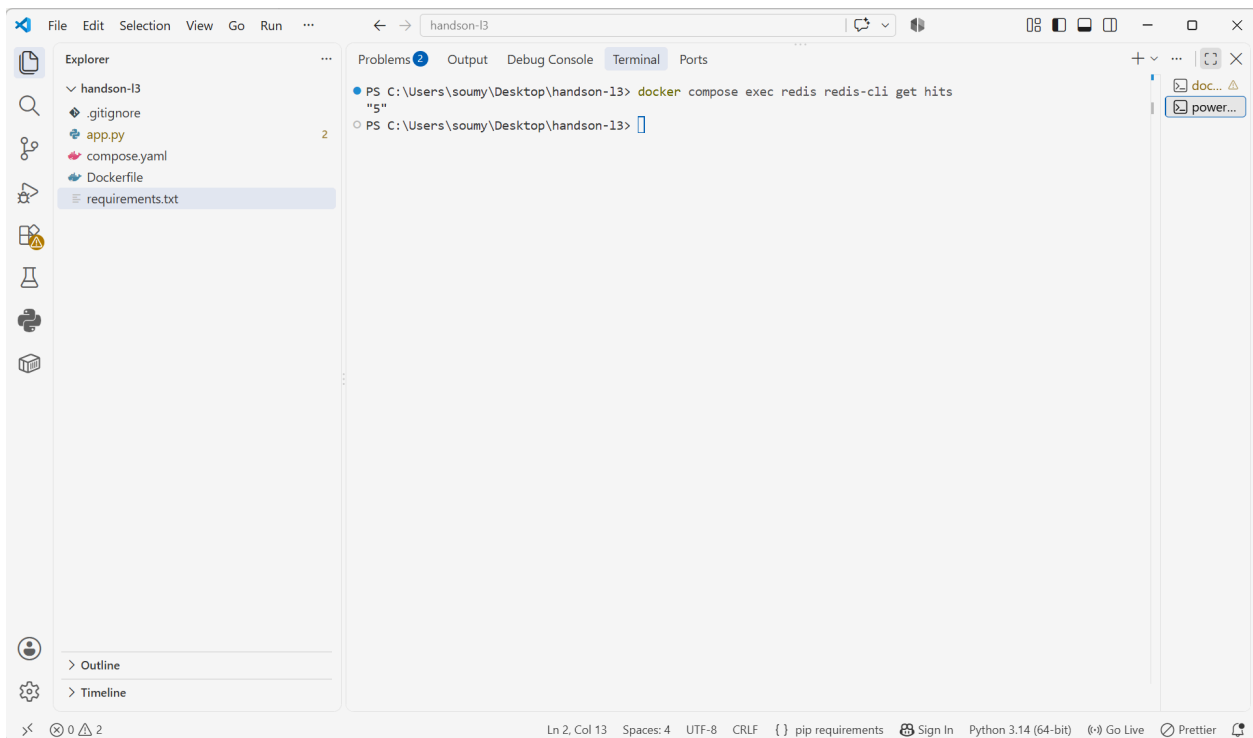


Figure 6 — Reading the counter directly from Redis with redis-cli, confirming the hit count is persisted in the cache rather than held in application memory.



Errors Logged

Four distinct errors were encountered and logged as GitHub Issues, each with the bug label and a closing comment describing the fix:

- 1. `SyntaxError: 'return' outside function` — `app.py`**
- 2. `Cannot connect to the Docker daemon (npipe)` — Windows**
- 3. `Docker Desktop failed to start: WSL not installed`**
- 4. `ModuleNotFoundError: No module named 'redis'` — running `app.py` on the host**